

Types of Variables — Answer Key

Ages 10-14 · Answers in bold. Accept any answer that means the same.

Independent variable (I) — the one factor you change on purpose.

Dependent variable (D) — the factor you measure to see the effect.

Controlled variable (C) — a factor you keep the same in every trial, so the test is fair.

Control group (G) — a separate group that gets no treatment, used as a baseline to compare against.

Part A — Sort the factors

Experiment: Does plant food make bean plants grow taller? Four groups of bean plants get 0 g, 2 g, 4 g or 6 g of plant food each week. Everything else is kept the same for all the plants. After three weeks, every plant's height is measured.

Write I, D, C or G in each box.

- I** 1. The amount of plant food each week
- D** 2. The height of the plants after three weeks
- C** 3. The amount of water each plant gets
- C** 4. The type of bean plant
- G** 5. The group given 0 g of plant food
- C** 6. The size of the pots
- C** 7. How much light the plants get
- C** 8. How long the test lasts (three weeks)

Note: the 0 g group is also one value of the independent variable.

Part B — Find the variables in each hypothesis

1. If the ramp is higher, then the toy car will roll farther.

Independent: height of the ramp **Dependent:** distance the car rolls

2. If a plant gets more hours of light each day, then it will grow more leaves.

Independent: hours of light each day **Dependent:** number of leaves

3. The warmer the water, the faster salt will dissolve in it.

Independent: temperature of the water **Dependent:** time to dissolve (or how fast)

4. If you exercise for longer, then your heart rate will be higher.

Independent: time spent exercising **Dependent:** heart rate (beats per minute)

Part C — Controlled variable or control group?

Circle the right answer for each statement.

1. Every pot gets exactly the same amount of water.

Controlled variable Control group

2. Some plants get no plant food, so the others can be compared with them.

Controlled variable **Control group**

3. All the plants are kept in the same room.

Controlled variable Control group